Tesselle

YOUR SHOPPING AGENT

GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Imagine it.
Keep what matters.
Get it made.

A personal shopping agent for things that have to fit. Gemini agents turn words, a photo, a video or a glance through smart glasses into a brief, find real stores and real makers that deliver to you, and keep your must-haves safe to delivery.

project-tesselle.web.app

Scene generated with Gemini



TEAM TESSELLE · ASHFAQUE RIFAYE

01 · THE PROBLEM

Inspiration is easy. Getting the right thing is not.

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Similar isn't suitable

A product that looks right can be 4 cm too deep, need drilling, or miss the one compartment that mattered.

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Search misses aren't answers

When stock doesn't fit, shoppers give up or quietly buy a compromise - and nobody can make the thing they actually wanted.

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Intent gets lost in the middle

Counteroffers, substitutions, quote changes and device switches rewrite requirements without anyone noticing.

The missing piece isn't more products. It's an agent that **preserves what the customer needs** through the messy commercial middle.

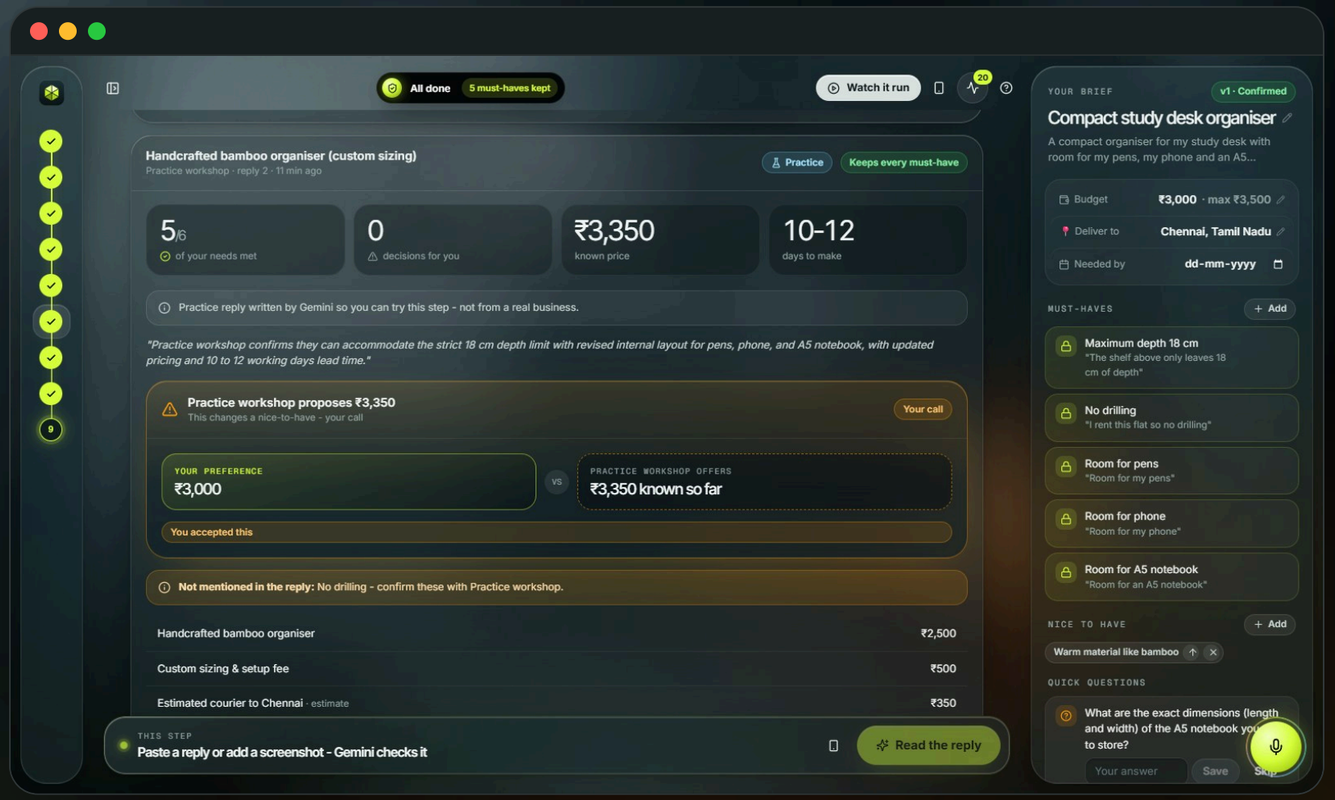
One brief. Real stores, real makers.

Your brief - kept safe

A living, versioned brief: must-haves, nice-to-haves, budget, delivery city and open questions. Edit it at any step, by typing or by voice. Nothing important changes without your decision.

Buy it now or get it made

Gemini with Google Search finds real listings (IKEA, Flipkart, Amazon.in, Pepperfry...) and real local makers, each linked to its page and checked against your must-haves. The agent writes the messages to makers; a counteroffer becomes your decision.

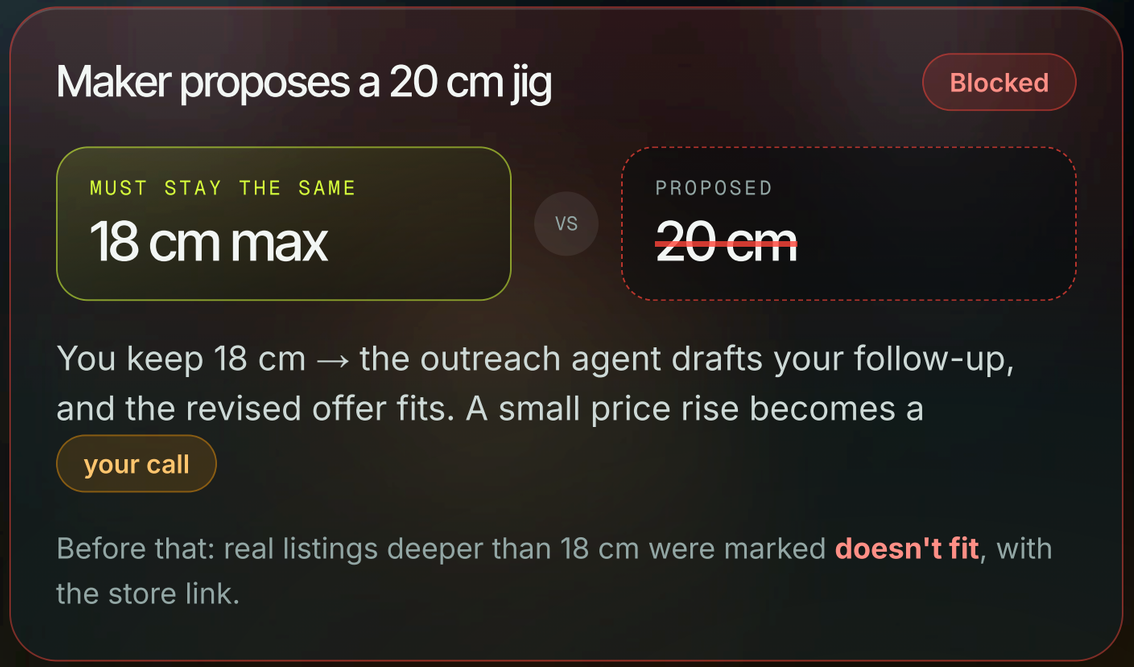


03 · HOW YOUR MUST-HAVES STAY SAFE

Language models propose. Deterministic code decides.

- **Fit verdicts** - fits / check a detail / doesn't fit, computed in code from structured requirements.
- **Quote deviations** - every offered spec is diffed against the brief version it answers.
- **Choose gates** - no stale quote, no changed brief, no unmet must-have, no double choice; anything unconfirmed needs a tick.
- **Honest uncertainty** - photo sizes stay "measure to confirm"; unknown charges never join a total.

Gemini agents only *explain* verdicts - a cheaper or prettier item that breaks a must-have is never promoted.

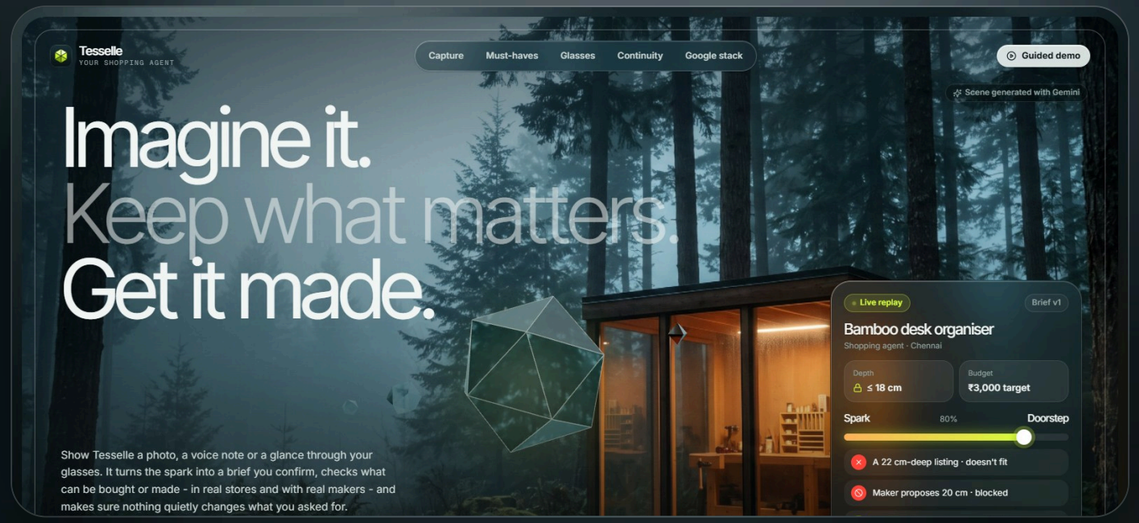


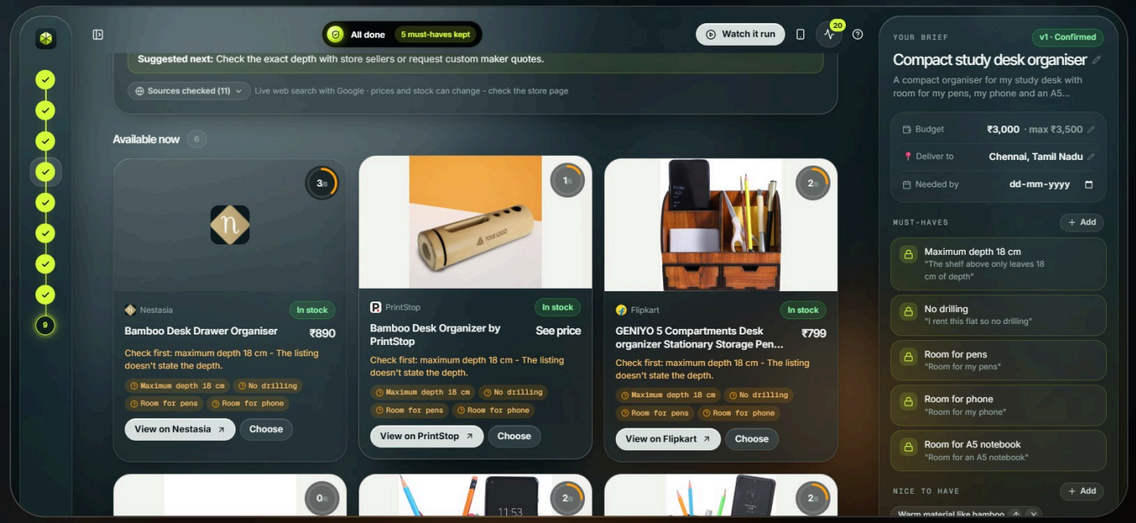
The full journey, tracked live on Google Cloud.

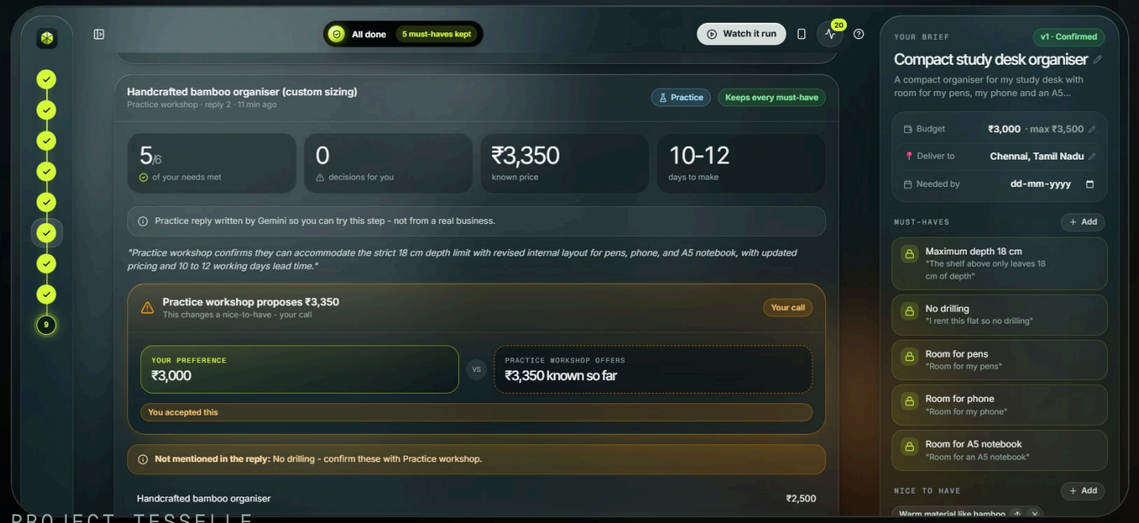
1. **Start**
   Talk or type + a photo: "18 cm shelf, I rent, no drilling, about ₹3,000"
2. **Your brief**
   The brief agent writes must-haves vs nice-to-haves; edit any time
3. **Visualise**
   Labelled concept; tap the picture to change a spot
4. **Options**
   Live web search: real listings + real makers near you, each checked
5. **Reach out**
   The agent writes the message; send it from Gmail or WhatsApp
6. **Replies**
   A 20 cm counteroffer is blocked until you decide
7. **Keep 18 cm**
   Follow-up drafted; the revised offer fits
8. **Any device**
   Phone app, voice assistant and glasses on the same project
9. **Choose & track**
   One option, one brief version; buy from the seller, mark it arrived
10. **Delivery check**
    Gemini vision checks the photo against the agreed checks

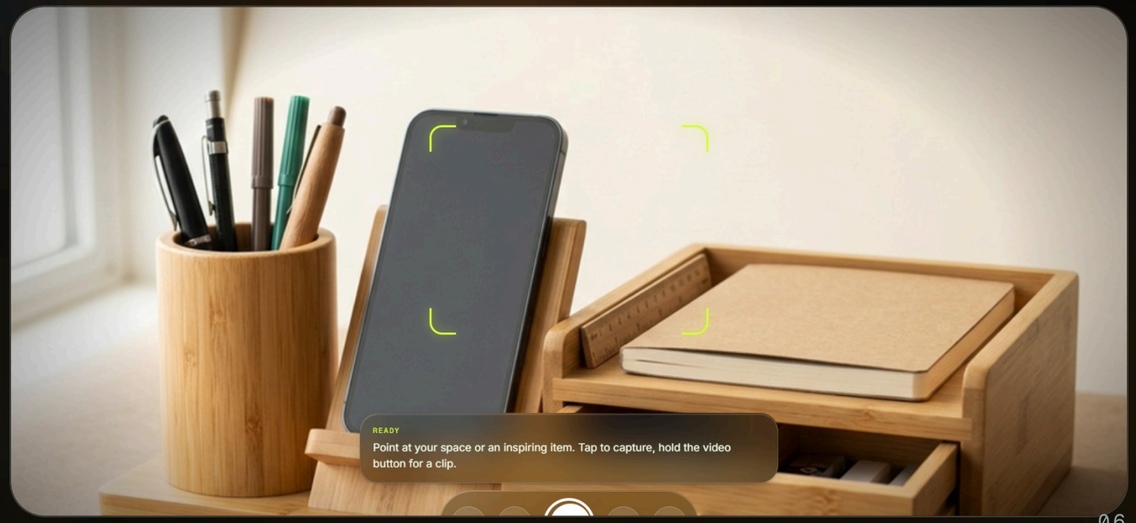
The guided demo runs this whole journey by itself, live; tools/e2e_storyboard.py asserts every guarantee.

Liquid glass, one next step - agents you can watch.









Gemini agents that act, grounded in the real web.

INTAKE

Multimodal intake

ADK LlmAgent on Gemini 3.8 Flash with typed output: text, voice, photos, **video** and generated images.

SEARCH

Shopping agent + maker scout

ADK 2 graph Workflow runs both in parallel → JoinNode → advisor. **Grounding with Google Search**; every result tied to its cited page; Gemini Embedding 2 ranks.

STUDIO

Creative studio

Art-director agent + Gemini image generation; click-to-pin visual prompting; flags edits that would touch the brief.

OUTREACH

Outreach & replies

Outreach agent writes quote requests and follow-ups; reply reader turns a pasted reply or screenshot into an offer; decision writer explains changes.

MEMORY

Memory & continuity

Agent Platform Sessions per project and **Memory Bank** for durable preferences ("I rent - no drilling").

VOICE + VISION

Voice & vision

Voice assistant with live speech-to-text and tools that edit the brief and drive the app; Gemini TTS replies; a delivery inspector that knows what a photo can't prove.

07 · ARCHITECTURE

Built end-to-end on Google Cloud.

SURFACES

React liquid-glass workspace + Three.js hero
Phone app (QR handoff, live sync)
Glasses HUD capture
PWA share target (Ray-Ban Meta)
Self-running guided demo

Firebase Hosting CDN

CLOUD RUN · FASTAPI + GOOGLE ADK 2

ADK LlmAgents with output schemas & tools
Graph Workflow (parallel search + join)
Fit checker - deterministic code
NDJSON progress streams with heartbeats
OpenTelemetry → Cloud Trace
Least-privilege service account · no API keys

VERTEX AI / AGENT PLATFORM

Gemini 3.8 Flash (text, vision, video)
Grounding with Google Search
Gemini 2.5 Flash (fast path + fallback)
Gemini 3.1 Flash Image
Gemini Embedding 2
Gemini 2.5 Flash TTS
Agent Runtime: Sessions + Memory Bank

FIRESTORE

Projects, immutable brief versions, options, messages, quotes, choices, activity

CLOUD STORAGE

Private media: inspirations, concepts, reply screenshots, delivery photos

CLOUD BUILD + REGISTRY

Container builds, asia-southeast1

LOGGING · TRACE · LABELS

Every Gemini call labelled app=tesselle; daily cost guard

08 · CAPTURE ANYWHERE

From a sentence to a glance through your glasses.

Speak it

Voice notes understood directly by Gemini.

Capture it

Live camera photos and short clips; video understanding.

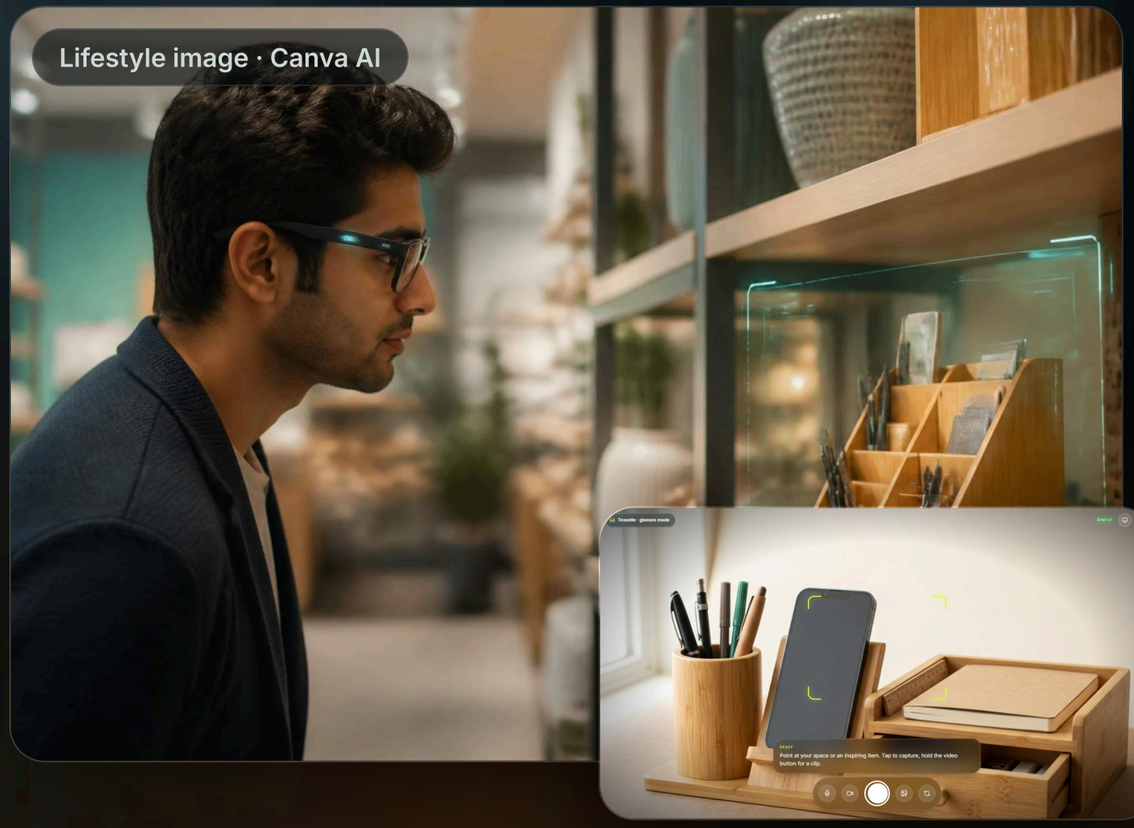
Imagine it

Gemini renders an inspiration, labelled as a concept.

Glance at it

Glasses HUD + PWA share target; replies spoken aloud.

Ray-Ban Meta captures sync to the phone and are shared straight to Tesselle. Direct camera streaming via Meta's native SDK is on the roadmap.



09 · IMPACT & BUSINESS

Unmet demand becomes actionable demand.

Customers

Less repeated searching and explaining, visible trade-offs, access to custom making - and confidence that what arrives is what they approved.

Makers & retailers

Qualified, structured briefs instead of vague enquiries; a way to serve demand existing stock can't. They can decline or propose - never be misrepresented.

Model

Disclosed fee on completed sourcing or custom orders; seller subscriptions for brief and quote management; optional design and sampling services.

What we'll validate (not claimed as achieved)

Brief completion · request-to-quote conversion · suitable-quote acceptance · requirement changes after approval · delivered-item acceptance - against an ordinary search-and-enquiry journey, with consented user testing.

10 · TRUST BY DESIGN

Honest by default. Safe by construction.

- Every listing links to its real page; prices and stock say "check the store page".
- Must-haves change only through a labelled decision that creates a new brief version.
- "No suitable match found in the sources checked" - an honest dead end beats a fabricated success.
- Concepts and the demo's practice reply are written by Gemini - and labelled so.

- Sellers receive only the brief - never your name, number, address or photos.
- Tesselle never pays, orders or sends messages for you; the assistant can't accept seller changes or choose for you.
- Vertex AI only, one service identity, no API keys; billing labels on every Gemini call.
- Public-demo cost guard caps daily model and image calls.

From one desk organiser to a demand engine.

NOW

Live prototype

Real listings and makers, voice assistant, phone + glasses capture, memory, a demo that runs itself.

NEXT

Agent sends for you

With your OK: Gmail and WhatsApp sending, reply inbox, maker verification.

THEN

Live voice & glasses

Gemini Live conversations; Meta wearables SDK for direct capture.

LATER

Shared demand

Opt-in pooling of compatible briefs when minimum quantities block a maker.

Tesselle

YOUR SHOPPING AGENT

TRY IT

Your intent, kept.

project-tesselle.web.app

tesselle-448771854429.asia-southeast1.run.app

Launch the guided demo - it runs itself, live, no sign-up. Real stores and makers; you stay in charge.



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